

Pinki Kumari

Second Year Graduate

Department of Computer Science and Engineering

pinkikumari22.github.io

Education

Degree	Institution	CPI	Year
MTech	IIT Gandhinagar	7.6	2022 - Present
BTech	VGI Microsoft, Greater Noida (AKTU)	8.33	2016 - 2020

Experience

- **Teaching Assistant, Indian Institute of Technology Gandhinagar:**
 - **Computing(ES112):** Designing and conducting lab tasks and checking assignments. [Aug 2023 - Present]
 - **Aarohan(FP601):** Managed orientation of 450+ students with a team of 25 TAs and was in the backend team, including five people. [July 2023]
 - **Data-Centric Computing:** Conducted labs and tutorials on topics of ML, EDA, Algorithms [Jan - July 2023]
 - **Computer Networks(CS433):** Handled a class of 75+ students, conducted quizzes, checked assignments, and assessed final project presentations. [Aug - Nov 2022]
- **Summer Intern, ThirdAI Corp.:** [June 1st - July 31st, 2023]
 - Utilized ThirdAI's Universal Deep Transformer (UDT) for pre-training, training, and fine-tuning LLMs on CPU instead of GPU.
 - Embedded evaluation system for benchmarking various models like BERT, Open AI, LLaMA, and BOLT.

Skill Summary

- **Languages:** Python, SQL, HTML5, CSS
- **Tools and Frameworks:** Kaggle, DrivenData, Git, Flask, VS Code
- **Data Science Libraries:** PyTorch, TensorFlow, Keras, Matplotlib, Seaborn, Scikit-Learn, Numpy, Pandas, JAX
- **General:** Machine Learning, Deep Learning, Natural Language Processing, Prompt Engineering

Projects

- **Hashing Algorithms for efficient training of LLMs and applying LSH on NTK:** [Jan 2023 - Present]
 - The objective is to probabilistically infer desired outputs without the need for explicit training of neural networks. This involves the construction of the NTK matrix(H^*).
 - Utilization of Locality Sensitive Hashing (LSH) in conjunction with Neural Tangent Kernels (NTK) under the guidance of [Prof. Anirban Dasgupta](#).
 - Applied concepts from Optimization Theory in Machine Learning and incorporated neural network principles.
- **Online Maintenance Complaint Portal:** [Jan - April 2023]
 - Designed a website for IIT Gandhinagar and created an Online Maintenance Complaint Portal that addressed existing challenges, including page reloading, authentication, and potential security vulnerabilities.
 - The task involved enhancing user experience and fortifying the website against potential threats.
 - Proficiently utilized SQL, Python, Flask, HTML5, and CSS.
- **Machine Learning:** [Jan - Oct 2023]
 - Trained a Named Entity Recognition (NER) model using spaCy to extract legal entities from legal documents. The trained model can recognize entities such as organizations, persons, dates and judges in legal texts.
 - Developed a Python library named "[customized-decision-tree-library-0.1](#)" for creating decision trees.
 - Created a Streamlit application to visualise the effects of cross-covariance by a 3-sigma ellipse plot.
- **Contact List Search:** [BTech Project]
 - Developed a user-friendly web application for contact management, featuring contact addition and deletion capabilities using buttons.
 - Implemented autocomplete functionality to enhance user experience, providing real-time contact name suggestions while typing.

Achievements

- Secured All India Rank (AIR) 61st (Armed Forces Quota) in Bihar Combined Entrance Competitive Exam (BCECE)
- Qualified for the GATE (CS) twice (2020, 2021)
- Secured 8th state rank in the International Olympiad of Mathematics 2010